

ENDOCRINE PHYSIOLOGY OBJECTIVES

HYPOTHALAMIC & PITUITARY HORMONES.

1. Understand the anatomical and functional endocrine relationships between the hypothalamus and pituitary gland (hypothalamic-pituitary axis).
2. Know the different communication pathways between the hypothalamus and anterior pituitary versus posterior pituitary lobes.
3. Know all hormones produced and/or secreted from the hypothalamus and their basic properties (Actions, Target Tissue, Hormone Class, Synthesis, Secretion, Storage, Transport, Metabolism, Mechanism of Actions, Regulation).
4. Know all hormones produced and/or secreted from the anterior pituitary and their basic properties.
5. Understand the basic pathophysiology and clinical consequences of any of the hypothalamic/pituitary disorders discussed.
6. Understand the basic negative feedback mechanism that regulates secretion of peripheral hormone under the control of the hypothalamic-anterior pituitary axis.
7. Understand the basic classification scheme of hypothalamic-ant. pituitary disorders (primary, secondary, tertiary) and be able to deduce the classification of specific and general disorders given appropriate laboratory or functional data.
8. Be able to differentiate above classification and predicted lab results scheme between hormones regulated by hypothalamic-anterior pituitary axis (hormone regulates hormone) and those that are not (e.g. insulin).
9. Know the basic hormone properties of growth hormone and be able to distinguish from other anterior pituitary hormones as well as other protein hormones?
10. Describe growth hormone's specific actions on energy metabolism and compare and contrast with those of catecholamines, glucocorticoids, thyroid hormones, insulin and glucagon.
11. Explain the regulatory relationship between GH and insulin-like growth factor I (IGF-I)
12. Understand the pathophysiology of the clinical manifestations resulting from GH excess and GH deficiency and be able classify as primary or secondary based on etiology, patient symptoms, and lab findings.
13. Be able to explain how pituitary diabetes develops in relation to GH metabolic actions.
14. Be able to describe the hypothalamic-pituitary feedback mechanisms that regulate GH secretion and know the major physiological factors that stimulate or inhibit GH secretion.
15. Understand how sleep influences GH secretion and how it differs from other sleep-regulated and circadian affected hormones such as prolactin and cortisol.
16. Be able to describe the general lifetime secretion pattern of growth hormone.
17. Know the basic hormone properties of prolactin and be able to compare and contrast with other anterior pituitary hormones as well as other protein hormones.
18. Describe the hypothalamic-pituitary feedback mechanisms that regulate prolactin secretion and know the major physiological factors that stimulate or inhibit GH secretion.
19. Understand how hypothalamic hormone regulation of prolactin differs from other anterior lobe hormones and be able to explain why interruption of the hypothalamic-pituitary portal circulation increases prolactin secretion relative to other anterior pituitary hormones

20. Explain open loop regulation in relation to prolactin regulation.
21. Understand how sleep influences prolactin secretion and how it differs from other sleep-regulated and circadian affected hormones presented later.
22. Know the major physiological actions of prolactin including the distinct role of prolactin during lactation relative to oxytocin.
23. Be able to predict the major pathophysiological manifestations expected during hyperprolactinemia including reproductive impairments in males as well as females.
24. Be able classify prolactin disorders as primary or secondary based on etiology, patient symptoms, and lab findings.
25. Know the basic hormone properties of the two posterior pituitary hormones and compare and contrast with anterior pituitary hormones as well as other protein hormones.
26. Describe the unique mechanism of posterior pituitary hormone biosynthesis and storage.
27. Know the physiological actions of ADH and oxytocin and the principle physiological regulators ADH and oxytocin secretion.
28. Know the feedback loops that regulate secretion of posterior pituitary hormones including positive feedback & open-loop regulation of oxytocin secretion.
29. Be able to describe the "Anticipatory Reflex" related to oxytocin's actions in milk ejection.
30. Be able to predict the relative effects of changes in plasma osmolality on ADH secretion, renal H₂O reabsorption, and urine osmolality.
31. Describe the pressor effect of vasopressin and be able to recognized clinical circumstances when ADH may exert this action.
32. Be able to predict the pathophysiological manifestation of ADH disorders based on functional and clinical laboratory information.